

CARSON SLATER

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EXPERIENCE

Research Assistant
Baylor University

January 2024 - Present
Waco, TX

- Developing statistical methods for urban water distribution processes, specifically with regard to of estimating outdoor watering behavior at the customer level and market segmentation.
- Collaborated with interdisciplinary domain experts to comprehend data and processes, perform robust analyses, and effectively communicate results.
- Developed R Shiny dashboard to enable collaborators at Denver Water to visualize the outdoor irrigation trends in homes with advanced metering infrastructure.
- Edited and reviewed material for an undergraduate intro to data science course. ([DOI: 10.7910/DVN/BZJW2E](https://doi.org/10.7910/DVN/BZJW2E))

Graduate Assistant
Baylor University

August 2023 - December 2025
Waco, TX

- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association.
- Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.
- Teaching assistant for Computational Statistics (R).

Academic Coach
Wheaton Academy Center for Lifelong Learning

January - May 2023
West Chicago, IL

- Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

Mathematics Teaching Assistant
Wheaton College

August 2022 - May 2023
Wheaton, IL

- Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.

SKILLS

Technology	Docker, Positron ($\approx$ VS Code), Git/Github, reveal.js, R/RStudio, SQL, duckDB, $\LaTeX$
Python	pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter
R	tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)
Stats & ML	statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods
Coursework	Statistical Bioinformatics, Deep Learning, Bayesian Methods, Bayesian Theory, Survival Analysis, Time Series, Theory of Statistics, Statistical Methods, Computational Statistics, Numerical Analysis, Linear Algebra for Data Analysis, Mathematical Statistics, Machine Learning, Probability Theory, Real Analysis, Intensive Introduction to Computer Science (Harvard), Econometrics, Economics of Labor and Poverty

EDUCATION

PhD Statistical Science, Baylor University

2024 - Pending

Research involves developing statistical methods for time series clustering with applications in urban water distribution processes.

M.S. Statistical Science, Baylor University
(GPA: 3.75/4.0)

2023 - 2024

B.S. Applied Mathematics with Statistics, Wheaton College

2019 - 2023

B.A. Economics

(GPA: 3.46/4.0), *Edman Presidential Scholarship*